

KNOWLEDGE REPRESENTATION

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CHAPTER-3

**Propositional and First-Order Logic: Representation, Inference,
Reasoning Patterns, and Resolution**

OutLine

PU

Introduction to Propositional Logic

And		
p	q	$p \cdot q$
T	T	T
T	F	F
F	T	F
F	F	F

Or		
p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

If ... then		
p	q	$p \supset q$
T	T	T
T	F	F
F	T	T
F	F	T

Not	
p	$\sim p$
T	F
F	T

What is Propositional Logic?

Propositional logic, also known as Boolean logic, deals with propositions which can be either true or false .

Examples of Propositions

- "It is raining."
- "The earth is flat."

Importance in Logic and Computer Science

Fundamental to the fields of logic, mathematics, and computer science, particularly in the development of algorithms and software verification.

Representation in Propositional Logic

Representation in Propositional Logic

Syntax

- Propositions: $(p, q, r, \dots)$

- Logical Connectives:

- Conjunction ($\wedge$): AND

- Disjunction ($\vee$): OR

- Negation ($\neg$): NOT

- Implication ($\rightarrow$): IF...THEN

- Biconditional ($\leftrightarrow$): IF AND ONLY IF

Cntd...

Semantics

- Truth Values: True (T) or False (F)
- Interpretation: Assignment of truth values to propositions
- Example:
 - $(p \wedge q) \rightarrow r$: If both (p) and (q) are true, then (r) must be true.

Inference in Propositional Logic



Definition: Logical inference is the process of deriving logical conclusions from given premises.

Methods:

- **Truth Tables:** Evaluate the truth value of propositions under all possible truth assignments.
- **Natural Deduction:** Use rules of inference to derive conclusions (e.g., Modus Ponens, Modus Tollens).
- **Example:**
 - Given: $(p \rightarrow q)$
 - Given: (p)
 - Infer: (q) (by Modus Ponens)

Reasoning Patterns in Propositional Logic

Reasoning Patterns in Propositional Logic

- Modus Ponens:

- If $(p \rightarrow q)$ and (p) is true, then (q) must be true.

- Example:

- **Premise:** $(\text{If } \text{it rains, the ground will be wet.})$

- **Premise:** It is raining.

- **Conclusion:** The ground is wet.

- Modus Tollens:

- If $(p \rightarrow q)$ and (q) is false, then (p) must be false.

Cont.....

Example:

- Premise: $\neg (\text{If } \neg)$ it rains, the ground will be wet.
- Premise: The ground is not wet.
- Conclusion: It is not raining.

Disjunctive Syllogism

- If $\neg (p \vee q)$ and $\neg (\neg p)$ is true, then $\neg (q)$ must be true.

Example:

- Premise: It is either raining or sunny.
- Premise: It is not raining.
- Conclusion: It is sunny.

Resolution in Propositional Logic

Resolution in Propositional Logic

Definition: Resolution is a rule of inference leading to a refutation proof by contradiction.

The Resolution Rule:

- Combine clauses to produce a new clause.
- Example:
 - From $\neg(p \vee q)$ and $\neg(\neg q \vee r)$, derive $\neg(p \vee r)$.
- **Example:**
 - Given: $\neg(p \vee q)$
 - Given: $\neg(\neg q \vee r)$
 - Resolution Step: $\neg(p \vee r)$

Complex Example in Propositional Logic

Complex Example in Propositional Logic

- Detailed Example:

- Premises:

- $(p \rightarrow q)$
- $(q \rightarrow r)$
- $(\neg r)$

- Truth Table Evaluation:

- (p) and (q) cannot both be true if $(\neg r)$ is true.

- Resolution:

- From $(\neg r)$, $(\neg q)$ (since $(q \rightarrow r)$)
- From $(\neg q)$, $(\neg p)$ (since $(p \rightarrow q)$)
- Conclusion: $(\neg p)$



Introduction to First-Order Logic (FOL)

Introduction to First-Order Logic (FOL).

Definition: First-order logic extends propositional logic by including quantifiers and predicates that allow statements about objects and their relationships.

- Differences from Propositional Logic:

- Involves variables, functions, and quantifiers (e.g., $\forall$ (forall), $\exists$ (exists)).
- Can express statements about multiple objects and their properties.

- Examples of FOL Statements:

- $\forall x (\text{Human}(x) \rightarrow \text{Mortal}(x))$: All humans are mortal.
- $\exists x (\text{Cat}(x) \wedge \text{Black}(x))$: There exists a black cat.

α'

$\vdots$

$\phi(\alpha')$

$\forall \alpha \phi(\alpha)$

Representation in First-Order Logic

Representation in First-Order Logic

Syntax

- **Variables**: $\{x, y, z, \dots\}$
- **Predicates**: $\{P(x), Q(x, y)\}$
- **Functions**: $\{f(x), g(x, y)\}$

Quantifiers

- Universal Quantifier ($\forall$): For all
- Existential Quantifier ($\exists$): There exists

Semantics

- **Domain of Discourse**: Set of all possible objects the variables can refer to.
- **Interpretation**: Assigns meaning to the symbols and formulas.

- Example:

- $\forall x (\text{Student}(x) \rightarrow \text{Enrolled}(x, \text{CS101}))$: All students are enrolled in CS101.

Inference in First-Order Logic

Inference in First-Order Logic

Definition: Logical inference in FOL involves deriving conclusions from a set of axioms using rules of inference.

-Methods:

-Unification: Process of finding a common instance of variables.

-Generalization: Inferring general rules from specific instances.

- Example:

- Given: $\forall x (\text{Human}(x) \rightarrow \text{Mortal}(x))$

- Given: $\text{Human}(\text{Socrates})$

- Infer: $\text{Mortal}(\text{Socrates})$ (using Universal Instantiation)

Reasoning Patterns in First-Order Logic

Reasoning Patterns in First-Order Logic

- Universal Instantiation:

- From $\forall x P(x)$, infer $P(a)$ for a specific a .

- Example:

- $\forall x (Bird(x) \rightarrow CanFly(x))$

- Infer: $Bird(Tweety) \rightarrow CanFly(Tweety)$

- Existential Instantiation:

- From $\exists x P(x)$, infer $P(c)$ for some c .

- Example:

- $\exists x (Cat(x) \wedge Black(x))$

- Infer: $Cat(Whiskers) \wedge Black(Whiskers)$

Resolution in First-Order Logic

Resolution in First-Order Logic

- **Definition:** Resolution in FOL is a rule of inference used for proving the unsatisfiability of a set of FOL clauses.
- **The Resolution Rule for FOL:**
 - Involves unifying pairs of clauses and resolving them.
- **Example:**
 - $\forall (\forall \text{forall } x (P(x) \vee Q(x)) \vee$
 - $\forall (\neg Q(a) \vee$
 - Unify $\forall (x \vee$ with $\forall (a \vee$, derive $\forall (P(a) \vee$
- **Example:**
 - Given: $\forall (P(f(x)) \vee Q(x) \vee$
 - Given: $\forall (\neg Q(a) \vee$
 - Unification and Resolution Step: $\forall (P(f(a)) \vee$

Advanced Example in First-Order Logic.

Advanced Example in First-Order Logic

- Detailed Example:

- Premises:

- $\forall x (P(x) \rightarrow Q(f(x)))$

- $P(a)$

- $\neg Q(f(a))$

- Unification and Resolution:

- From $P(a) \rightarrow Q(f(a))$



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